How to factorize an expression

Factorizing an expression is the opposite of expanding an expression. To do this, look for a factor (number or letter) that all the terms (parts) of the expression have in common. The common factor can then be placed outside a parenthesis enclosing what is left of the other terms.

4 is common to both 4b and 12 (because they can both be divided by 4)

$$4b + 12$$

this means $4 \times b$

To factorize an expression, look for any letter or number (factor) that all its parts have in common.

this is the same as 12

$$4 \times b + 4 \times 3$$

both b and 3 are not common to both parts so they go inside the parenthesis

In this case, 4 is a common factor of both 4b and 12, because both can be divided by 4. Divide each by 4 to find the remaining factors of each part. These go inside the parenthesis.

place 4 outside parenthesis

remaining factors go inside parenthesis

$$4(b + 3)$$

parenthesis means multiply

Simplify the expression by placing the common factor (4) outside a parenthesis. The other two factors are placed inside the parenthesis.

Factorizing more complex expressions

Factorizing can make it simpler to understand and write complex expressions with many terms. Find the factors that all parts of the expression have in common.

3×3 | $3 \times 3 \times x \times x \times y = 9x^2y$ | $x \times x$ | 3×5 | $3 \times 5 \times x \times x \times y \times y = 15xy^2$ | $x \times y^2$ | $y \times y$ | $2 \times 3 \times 3 \times x \times x \times y \times y \times y = 18xy^3$

$$9x^2y + 15xy^2 + 18xy^3$$

all 3 terms multiplied

To factorize an expression write out the factors of each part, for example, y^2 is $y \times y$. Look for the numbers and letters that are common to all the factors.

common factor of x variables

common factor of numbers

$$3xy$$

common factor of y variables

All the parts of the expressions contain the letters x and y, and can be factorized by the number 3. These factors are combined to produce one common factor.

3xy is common factor of all parts of the expression

$9x^2y \div 3xy = 3x$

$15xy^2 \div 3xy = 5y$

$18xy^3 \div 3xy = 6y^2$

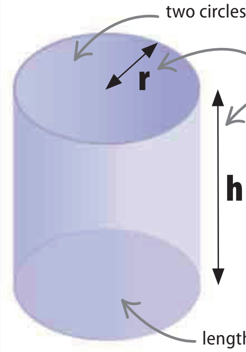
$$3xy(3x + 5y + 6y^2)$$

Set the common factor (3xy) outside a set of parentheses. Inside, write what remains of each part when divided by it.

LOOKING CLOSER

Factorizing a formula

The formula for finding the surface area (see pp.156–157) of a shape can be worked out using known formulas for the areas of its parts. The formula can look daunting, but it can be made much easier to use by factorizing it.



◁ **Surface of a cylinder**
The formula for the surface area of a cylinder is worked out by adding together the areas of the circles at each end and the rectangle that forms the space between them.

area of rectangle is length (2πr) height (h)

$$2\pi rh + 2\pi r^2$$

area of a circle is πr^2, for 2 circles it is $2\pi r^2$

To find the formula for the surface area of a cylinder, add together the formulas for the areas of its parts.

2πr is common to both expressions

means multiply by

h and r are not common to both terms so they sit inside the parenthesis

$$2\pi r(h + r)$$

To make the formula easier to use, simplify it by identifying the common factor, in this case 2∏r, and setting it outside the parentheses.